PN Junction - Current

Chapter 5:

PN Junctions

•Electrostatics

•Current

Streetman and Banerjee, 'Solid State Electronic Devices,' 6th ed., Pearson, 2006.

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Learning Objectives

Item	OUTCOME-RELATED LEARNING OBJECTIVES	OUTCOMES													
		1A	1B	1C	2	3	4	5	6	7					
1	Ability to determine crystallographic directions, planes, equivalent directions, and equivalent planes.		x	x											
2	Understand an energy band diagram including Ec, Ev, Ei, Ef, Fn, and Fp.		x												
3	Understand the concepts of conductivity and resistivity in terms of carrier concentration and mobility; ability to calculate sheet resistance.		x												
4	Ability to determine the built-in voltage, depletion width, junction capacitance, and current of an ideal pn junction diode.	x	x												
5	Understand temperature dependence of the current of a pn junction.														
6	Understand the effects of non-idealities in a pn junction: R-G, high level injection, ohmic drop, avalanche and Zener breakdown	x	x												
7	Basic understanding of Schottky diodes and thermionic emission.	x	x												
8	Ability to extract flat-band voltage, threshold voltage, oxide capacitance, & depletion capacitance from a MOS CV.	x	x												
9	Understand MOSFET short channel effects and the inverse subthreshold slope.	x	x												

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PN Junction Current

- Intuitive picture of diode current flow
- Quantitative calculation of diode current
  - Use minority carrier diffusion equations (MCDEs) to derive the Ideal Diode Equation
  - Non-idealities

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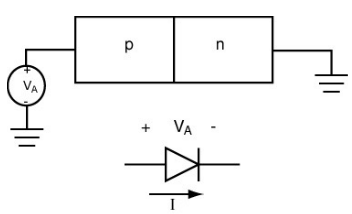
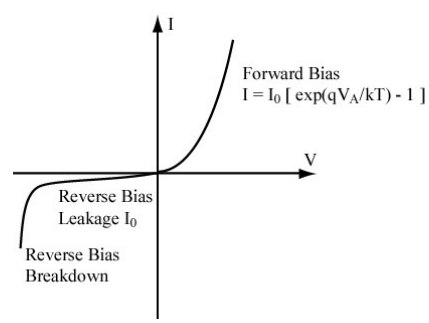
Idealized PN Junction

- Current – Voltage response.
  - Derive $I = I_0 [\exp(qV_A/kT) - 1]$



I-V Curve



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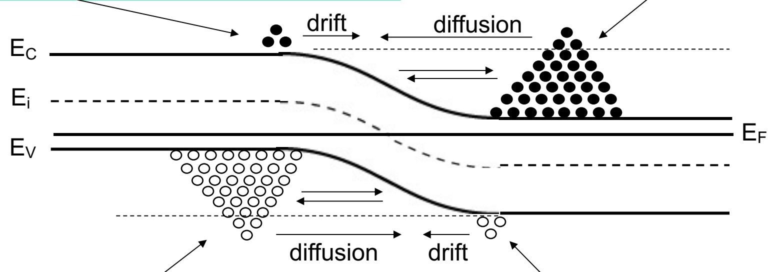
Energy Distribution of Carriers in **Equilibrium**.

These few minority electrons near the depletion region fall down the hill and are swept across to the n-side. This is the <u>drift current</u>, and it is limited by thermal generation of minority carriers ($g_i = \alpha_r n_o p_o$).

These majority carriers near the depletion region diffuse towards the junction. Only the ones with enough kinetic energy can get across.



These majority carriers near the depletion region diffuse towards the junction. Only the ones with enough kinetic energy can get across.

These few minority holes near the depletion region fall down the hill and are swept across to the n-side. This is the <u>drift current</u>, and it is limited by thermal generation of minority carriers.

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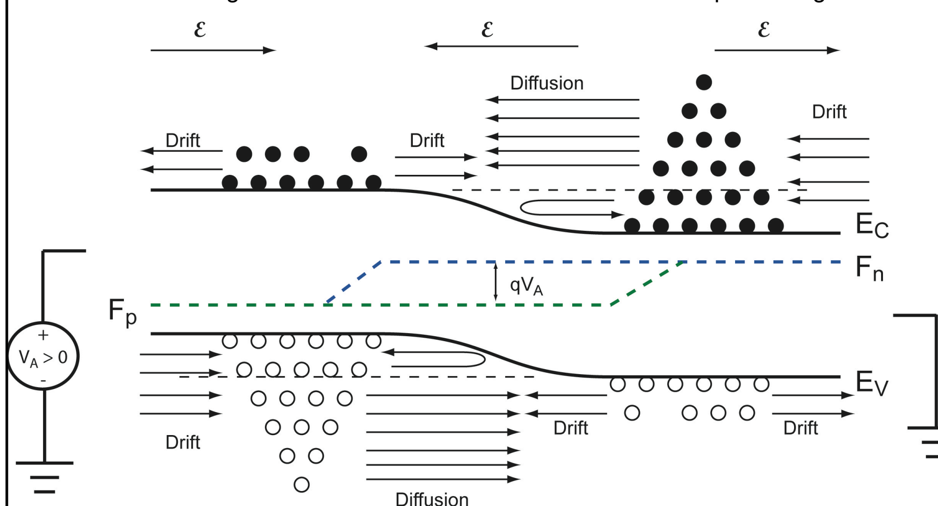
Energy Distribution of Carriers in **Forward Bias.**

- Forward bias reduces the barrier (qV_{bi}) in the junction to $q(V_{bi} – V_A)$.
- Note the change in the direction of the electric field in the depletion region.



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Energy Distribution of Carriers in **Forward Bias**.

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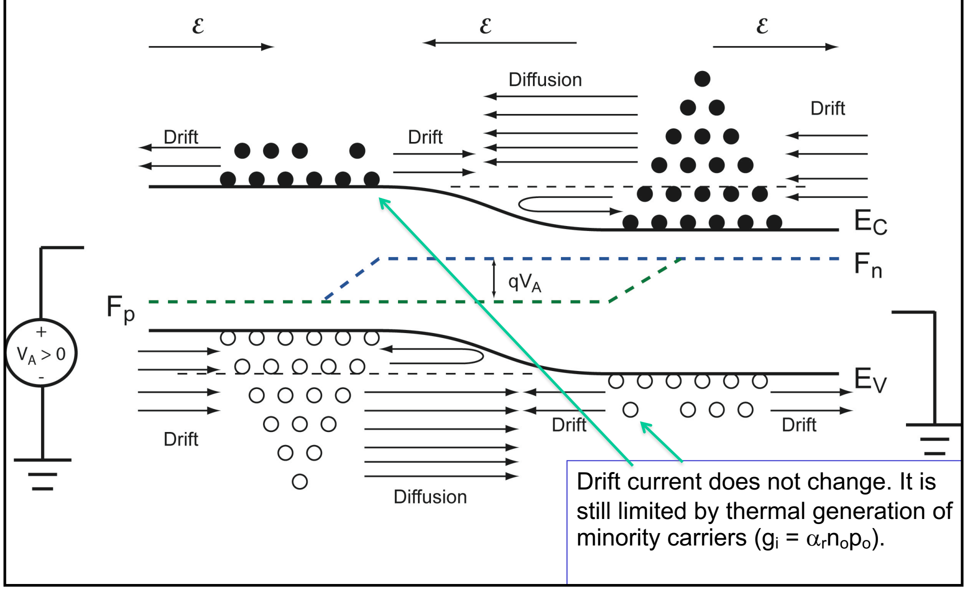

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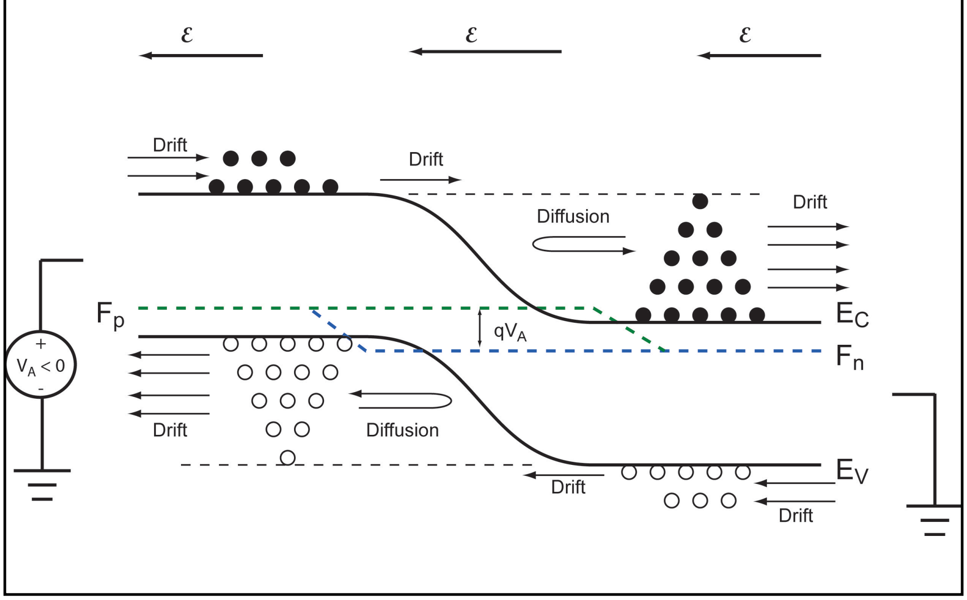
Cartoon of the Energy Distribution of Carriers in **Reverse Bias**

- Reverse bias increases the barrier in the junction..



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IDEAL PN Junction Current – Notes and Assumptions

1. V_A is applied to the p-side. Positive V_A lowers the barrier of qV_{Bi}.
2. Voltage drop is predominantly in the space-charge (depletion) region.
3. Total current is conserved so that it can be evaluated at any point in the device.
4. Low level injection applies: $\delta n_n << n_{n,0}$ and $\delta p_p << p_{p,0}$. i.e. Majority carrier populations are negligibly changed at the edges of the depletion region. $n(x_n) \approx N_D$ and $p(-x_p) \approx N_A$.
5. No R-G in the depletion region. No light.
6. Depletion approximation and step junction
7. Steady State (D.C.).

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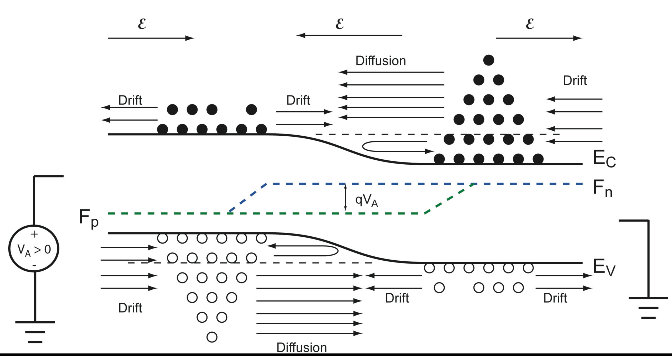
IDEAL PN Junction Current – Notes and Assumptions

8. Drift Current is insensitive to $(V_{bi} – V_A)$.
   1. It is insensitive to the height of the potential barrier or the electric field in the depletion region.
   2. Limited by thermal generation of minority carriers near the junction.
   3. Referred to as generation current.



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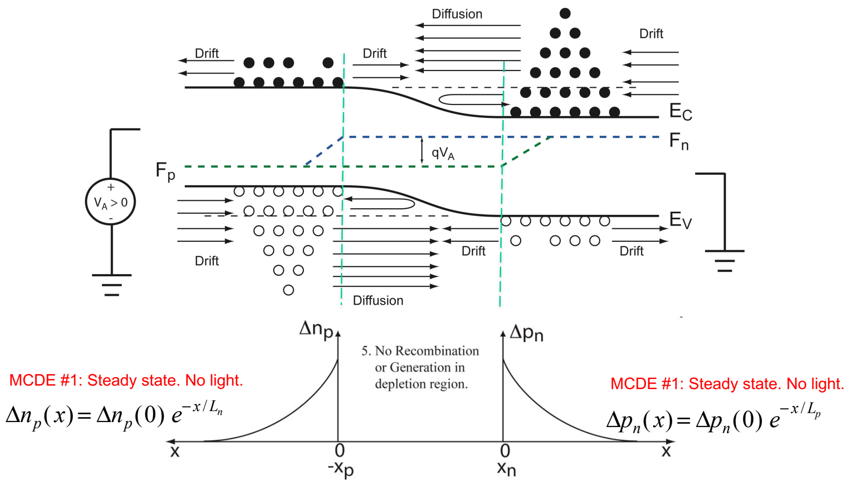
Approach to Derive the Ideal Diode Equation

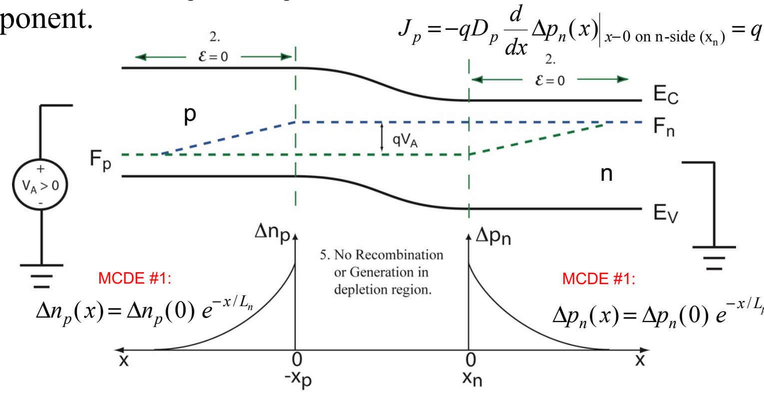


- Determine $J_{n,Diff}(-x_p)$ and $J_{p,Diff}(x_n)$ and add. No R-G in depletion region ➔ $J_n(-x_p) = J_n(x_n)$.
- Subtract off equilibrium component.
- The equilibrium component is the drift component shown.

Derive the Ideal Diode Equation

- Determine $J_{n,Diff}(-x_p)$ and $J_{p,Diff}(x_n)$ and add. Subtract off equilibrium component.

$$J_p = -qD_p \frac{d}{dx}\Delta p_n(x)\Big|_{x=0 \text{ on n-side } (x_n)} = q\frac{D_p}{L_p}\Delta p_n(0$$

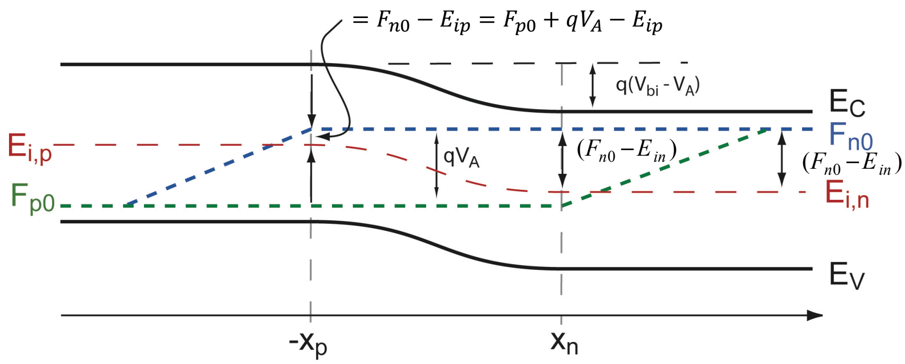


$$J_n = -qD_n \frac{d}{dx}\Delta n_p(x)\Bigg|_{x=0 \text{ on p-side } (-x_p)} = q\frac{D_n}{L_n}\Delta n_p(0)$$

$$J_p = -qD_p \frac{d}{dx}\Delta p_n(x)\Bigg|_{x=0 \text{ on n-side } (x_n)} = q\frac{D_p}{L_p}\Delta p_n(0)$$

The diffusion current calculated from Δn and Δp automatically subtracts off the equilibrium current. Need $\Delta n_p(0)$ and $\Delta p_n(0)$.

$\Delta n_p(0)$ and $\Delta p_n(0)$



$$n_p(-x_p) = n_i e^{(F_{n0}-E_{ip})/kT} = n_i e^{(F_{p0}+qV_A-E_{ip})/kT} = n_{p0} e^{qV_A/kT}$$

$$n_p(-x_p) = n_{p0} e^{qV_A/kT} \implies \Delta n_p(-x_p) = n_{p0} e^{qV_A/kT} - n_{p0}$$

$$\implies \boxed{\Delta n_p(-x_p) = n_{p0}\left(e^{qV_A/kT} - 1\right)}$$

Similarly, $\boxed{\Delta p_n(x_n) = p_{n0}\left(e^{qV_A/kT} - 1\right)}$

These are the boundary conditions that we need.

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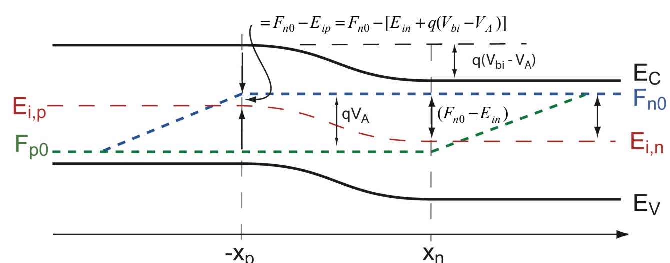
Law of the Junction and $\Delta n_p(0)$ and $\Delta p_n(0)$



Anywhere in the depletion region:

$$n = n_i e^{(F_{n0}-E_i)/kT} \text{ and } p = n_i e^{(E_i-F_{p0})/kT}$$

$$np = n_i^2 e^{(F_{n0}-F_{p0})/kT} = n_i^2 e^{qV_A/kT}$$

Law of the junction: Anywhere in the depletion region, $np = n_i^2 e^{qV_A/kT}$

At the depletion region edges, $p(-x_p) = N_A$ and $n(x_n) = N_D$ (assumption 4), therefore,

$$n(-x_p) = \frac{n_i^2}{N_A} e^{qV_A/kT} = n_{p0} e^{qV_A/kT}$$

$$p(x_n) = \frac{n_i^2}{N_D} e^{qV_A/kT} = p_{n0} e^{qV_A/kT}$$

$$\implies \boxed{\begin{aligned} \Delta n_p(-x_p) &= n_{p0}\left(e^{qV_A/kT} - 1\right) \\ \Delta p_n(x_n) &= p_{n0}\left(e^{qV_A/kT} - 1\right) \end{aligned}}$$

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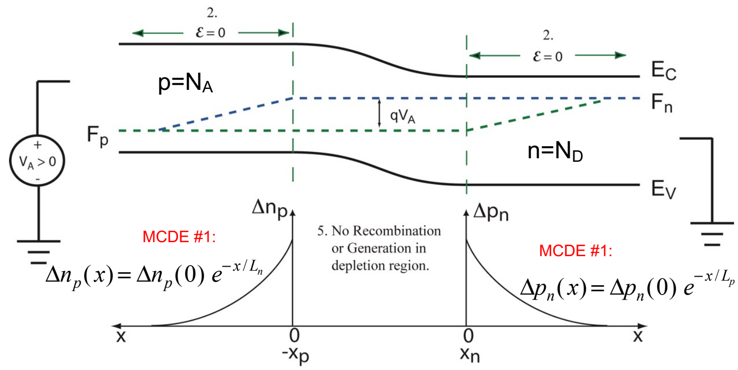
Evaluate the Current



$$J_n=-qD_n\frac{d}{dx}\Delta n_p(x)\Big|_{x=0\text{ on p-side }(-x_p)}=q\frac{D_n}{L_n}\frac{n_i^2}{N_A}\left(e^{qV_A/kT}-1\right)$$

$$J_p=-qD_p\frac{d}{dx}\Delta p_n(x)\Big|_{x=0\text{ on n-side }(x_n)}=q\frac{D_p}{L_p}\frac{n_i^2}{N_D}\left(e^{qV_A/kT}-1\right)$$

$$\boxed{J=q\,n_i^2\left[\frac{D_n}{N_A\,L_n}+\frac{D_p}{N_D\,L_p}\right]\left(e^{qV_A/kT}-1\right)}$$

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Ideal Diode Current

- Several ways to write it:

$$\boxed{J=q\,n_i^2\left[\frac{D_n}{N_A\,L_n}+\frac{D_p}{N_D\,L_p}\right]\left(e^{qV_A/kT}-1\right)}$$

$L_p=\sqrt{D_p\tau_p}$ or $D_p=L_p^2/\tau_p$

$L_n=\sqrt{D_n\tau_n}$ or $D_n=L_n^2/\tau_n$

$$\boxed{J=q\,n_i^2\left[\frac{1}{N_A}\sqrt{\frac{D_n}{\tau_n}}+\frac{1}{N_D}\sqrt{\frac{D_p}{\tau_p}}\right]\left(e^{qV_A/kT}-1\right)}$$

$$\boxed{J=q\left[n_{p0}\frac{L_n}{\tau_n}+p_{n0}\frac{L_p}{\tau_p}\right]\left(e^{qV_A/kT}-1\right)}$$

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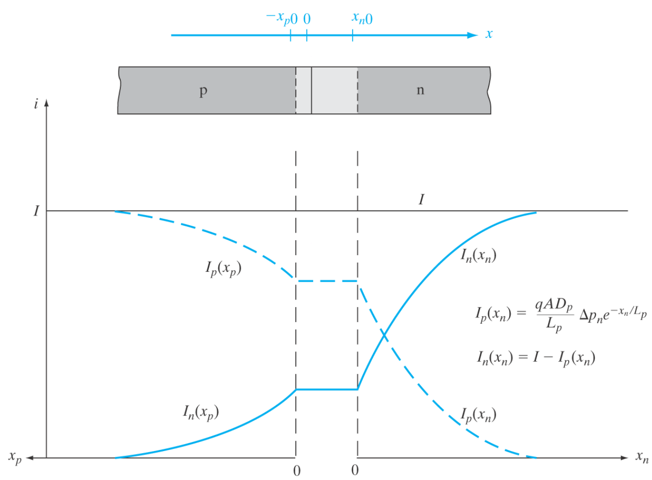
Electron and Hole Current throughout the Device



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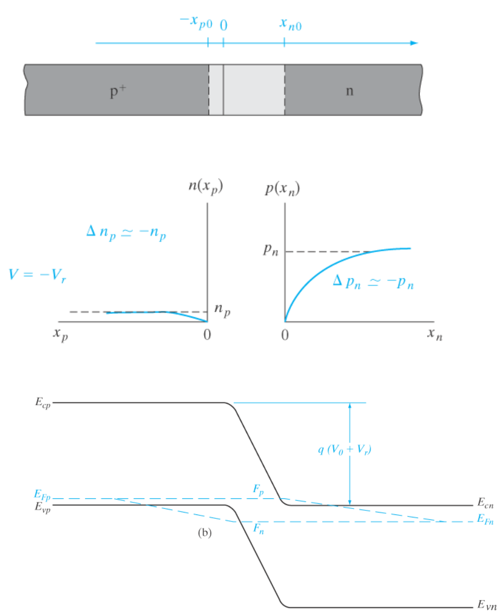
Reverse Bias Current



$|V_A| >> kT$:

$$\Delta n_p(-x_p) = \frac{n_i^2}{N_A}\left(\cancelto{0}{e^{qV_A/kT}} - 1\right)$$

$$\Delta p_n(x_n) = \frac{n_i^2}{N_D}\left(\cancelto{0}{e^{qV_A/kT}} - 1\right)$$

There are NO minority carriers at the edges of the depletion regions.

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Ideal Reverse Biased Diode Current

- Several ways to write it:

$$J = q\, n_i^2 \left[\frac{D_n}{N_A\, L_n} + \frac{D_p}{N_D\, L_p} \right] \left(\xcancel{e^{qV_A/kT}}^{\,0} - 1 \right)$$

$|V_A| >> kT$:

$L_p = \sqrt{D_p \tau_p}$ or $D_p = L_p^2 / \tau_p$

$L_n = \sqrt{D_n \tau_n}$ or $D_n = L_n^2 / \tau_n$

$$J = q\, n_i^2 \left[\frac{1}{N_A} \sqrt{\frac{D_n}{\tau_n}} + \frac{1}{N_D} \sqrt{\frac{D_p}{\tau_p}} \right] \left(\xcancel{e^{qV_A/kT}}^{\,0} - 1 \right)$$

$$J = q \left[n_{p0} \frac{L_n}{\tau_n} + p_{n0} \frac{L_p}{\tau_p} \right] \left(\xcancel{e^{qV_A/kT}}^{\,0} - 1 \right)$$

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Ideal Reverse Biased Diode Current

- Several ways to write it:

$$J = -q\, n_i^2 \left[\frac{D_n}{N_A\, L_n} + \frac{D_p}{N_D\, L_p} \right]$$

$|V_A| >> kT$:

$L_p = \sqrt{D_p \tau_p}$ or $D_p = L_p^2 / \tau_p$

$L_n = \sqrt{D_n \tau_n}$ or $D_n = L_n^2 / \tau_n$

$$J = -q\, n_i^2 \left[\frac{1}{N_A} \sqrt{\frac{D_n}{\tau_n}} + \frac{1}{N_D} \sqrt{\frac{D_p}{\tau_p}} \right]$$

$$J = -q \left[n_{p0} \frac{L_n}{\tau_n} + p_{n0} \frac{L_p}{\tau_p} \right]$$

$$J = -I_0$$

In general:

$$I_0 = q \left[n_{p0} \frac{D_n}{L_n} + p_{n0} \frac{D_p}{N_D\, L_p} \right]$$

$$I = I_0 \left(e^{qV_A/kT} - 1 \right)$$

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Temperature Dependence of Ideal Diode Current

***ABET CO 5**: Understand temperature dependence of the current of a pn junction.*

$$J = qn_i^2\left[\frac{D_n}{N_A L_n}+\frac{D_p}{N_D L_p}\right]\left(e^{qV_A/(kT)}-1\right) = I_0(T)\left(e^{qV_A/(kT)}-1\right)$$

$$n_i^2 = N_C N_V e^{-E_G/(k_BT)} = 4\left(m_n^* m_p^*\right)^{3/2}\left(\frac{k_B}{2\pi\hbar^2}\right)^3 T^3 e^{-E_G/(k_BT)}$$

$$n_i^2 \propto T^3 e^{-E_G/(k_BT)}$$

Neglecting the temperature dependence of D_n/L_n and D_p/L_p, we have,

$J \propto T^3 e^{-(E_G-qV_A)/(k_BT)}$ $\quad (3k_BT < V_A < E_G)$ Forward bias

$J \propto T^3 e^{-E_G/(k_BT)}$ $\quad (V_A < -3k_BT)$ Reverse bias

Circuit Design Consideration: Consider a Si diode biased at $V_A = 0.72$ V. For automotive applications, integrated circuits must operate in the temperature range of -40º C to +125º C. Assume that the diffusion coefficients, lifetimes, and bandgap are unaffected by the temperature change. If the applied voltage is fixed at $V_A = 0.72$ V, what is the ratio of the max current to the min current over this temperature range?
$T_L = 233\ K$ and $T_H = 398K$. Then, $kT_L = 0.026 * \frac{233}{300} = 0.0202$ eV and $kT_H = 0.026 * \frac{398}{300} = 0.0345$ eV.

$$\frac{J(T_H)}{J(T_L)} = \left(\frac{T_H}{T_L}\right)^3 \frac{e^{-(E_G-qV_A)/kT_H}}{e^{-(E_G-qV_A)/kT_L}} = \left(\frac{T_H}{T_L}\right)^3 exp\left[-(E_G-qV_A)\left(\frac{1}{kT_H}-\frac{1}{kT_L}\right)\right] = \left(\frac{398}{233}\right)^3 \exp[(0.4)(20.519)]$$
$$= 1.828\times10^4$$

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Temperature Dependence of Ideal Diode Current

Research Application: Consider a pn diode built from a new semiconductor. How do you estimate the electronic bandgap? Plot ln(J) vs. $1/k_BT$: (**Arrhenius plot**). Let $V_A = 0.72$ V.

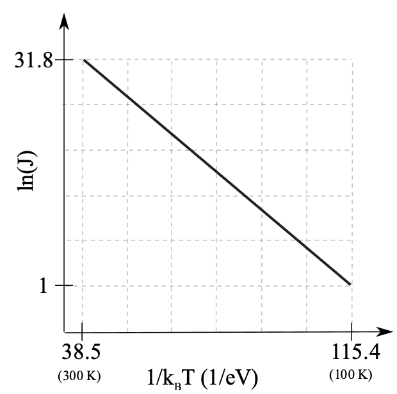
$\ln(J) = \ln(\text{constant}) + 3\ln(T) - \frac{(E_G - qV_A)}{k_BT}$ $\quad (3k_BT < V_A < E_G)$ Forward bias

$\ln(J) = \ln(\text{constant}) + 3\ln(T) - \frac{E_G}{k_BT}$ $\quad (V_A < -3k_BT)$ Reverse bias

Ignore $\ln(T)$ term. The slope under forward bias is $-(E_G - qV_A)$. Therefore,

$$\frac{1-31.8}{115.4-38.5} = -0.4\ eV = -(E_G - qV_A)$$

$$E_G = 0.4 + qV_A\ \text{eV} = 1.12\ \text{eV}.$$



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Non-Idealities Sec. (5.6)

ABET CO 6: Understand the effects of non-idealities in a pn junction: R-G, high level injection, ohmic drop, avalanche and Zener breakdown

$$I = I_0\left(e^{qV_A/nkT} - 1\right)$$

Ideality factor

- See what this ‘ideality factor’ is all about.
- Relax some of the assumptions that went into the derivation of the ideal diode equation.

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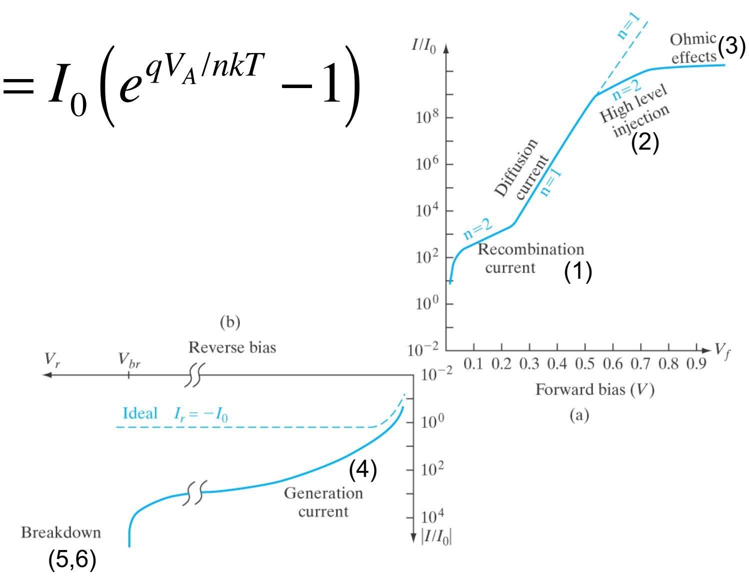
Non-Idealities Sec. (5.6)

$$I = I_0\left(e^{qV_A/nkT} - 1\right)$$



From *Solid State Electronic Devices,* Sixth Edition, by Ben G. Streetman and Sanjay Kumar Banerjee.

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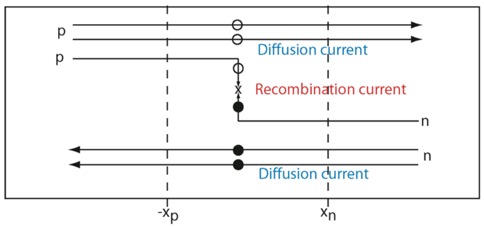
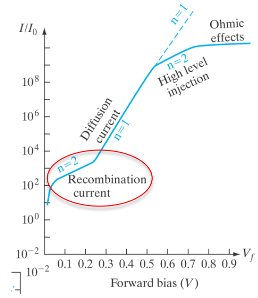
Forward Bias Non-Idealities Sec. (5.6)

1. Recombination in the junction under forward bias. Sec. (5.6.2)

- Allow recombination in the junction (violating assumption 5).





- Additional current, Recombination current: $I_R \propto n_i\left(e^{qV_A/2kT}-1\right)$
- Diffusion current: $I_D \propto n_i^2\left(e^{qV_A/kT}-1\right)$
- For small n_i (wide band gap such as Si and GaAs), recombination in the depletion region dominates at low V_A.

$$I \propto n_i\left(e^{qV_A/2kT}-1\right)$$ at low V_A.

$$n_i = \sqrt{N_C N_V}\, e^{-E_G/2kT}$$

n=2 at low forward bias for wider bandgap semiconductors
n=1 for lower bandgap semiconductors like Ge & InAs.

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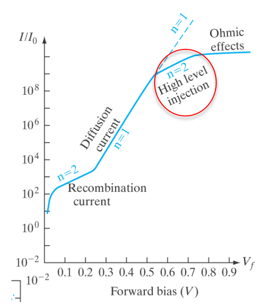
Forward Bias Non-Idealities

2. High level injection (opposite of low-level injection): $\Delta n_n > n_{n,0}$ (violating assumption 4) Sec. (5.6.1).



- Large forward bias: MAJORITY CARRIER concentration is perturbed: $n_n(x_n) \approx \Delta n_n = \Delta p_n$
- Using the law of the junction, anywhere in the depletion region, $np = n_i^2 e^{qV_A/kT}$:

$$n(x_n)p(x_n) = \left(n_{n0}+\Delta n_n\right)\left(p_{n0}+\Delta p_n\right) \approx \left(\Delta p\right)^2 = n_i^2 e^{qV_A/kT}$$

$$\Delta p(x_n) = n_i e^{qV_A/2kT}$$

$$I \propto n_i e^{qV_A/2kT}$$ at high V_A.

n=2 at high forward bias due to HIGH LEVEL INJECTION

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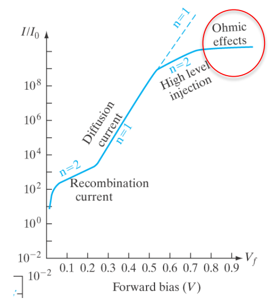
Forward Bias Non-Idealities

3. Ohmic Effects Sec. (5.6.3)



- Under large forward bias (very high currrent), the resistance of the n and p regions can become important. This resistance causes some of the applied voltage to be dropped in the n and p regions, and NOT across the junction (violating assumption 2).
- If V_A is the applied voltage at the contacts, the voltage across the junction is:

$$V_A' = V_A - RI = V_A - RI_0 e^{qV_A'/2kT}$$

- As the current becomes limited by the resistance, the I-V response becomes more linear and flattens out on a semi-log plot.

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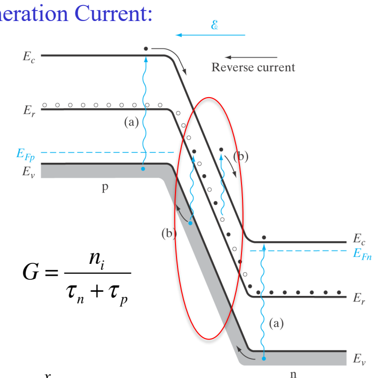
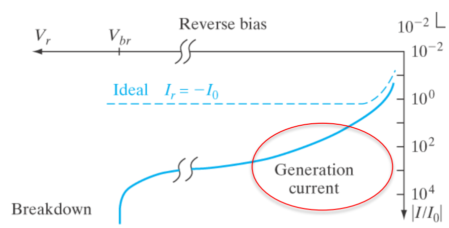
Reverse Bias Non-Idealities

4. Generation in the depletion region (violating assumption 5) Sec. (5.6.2)

- Generation Current:



$$G = \frac{n_i}{\tau_n + \tau_p}$$



$$I_G = -q \int_{-x_p}^{x_n} dx\, G = -q \frac{n_i}{\tau_n + \tau_p} W$$

$$I_r = -I_0 \underbrace{- q \frac{n_i}{\tau_n + \tau_p} W}_{I_G}$$

$$I_0 \propto n_i^2 \qquad I_G \propto n_i$$

For large bandgap (n_i small), I_G dominates.
For small bandgap (n_i large) I_0 dominates. The reverse bias current will look 'ideal.'

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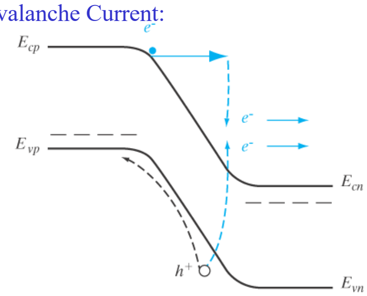
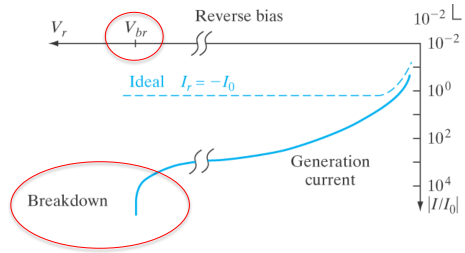
Reverse Bias Non-Idealities

5. Avalanche Current Breakdown (violating assumption 5) Sec. (5.4.2)

- Avalanche Current:





- An electron or hole with kinetic energy > E_G can excite an electron-hole pair in the high-field, depletion region.
- The new EHPs can gain kinetic energy > E_G and excite another set of EHPs. This is positive feedback which can result in an avalanche effect and breakdown.
- Multiplication factor M: $M = \dfrac{1}{1-\left(\dfrac{|V_A|}{V_{Br}}\right)^m}$ $\quad (3 \le m \le 6)$

$$V_{br} \propto \left(\frac{1}{N_A} + \frac{1}{N_D}\right)$$

p^+-n: $V_{br} \sim 1/N_D$
n^+-p: $V_{br} \sim 1/N_A$

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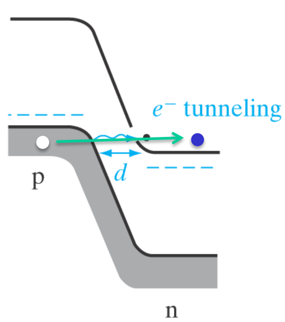
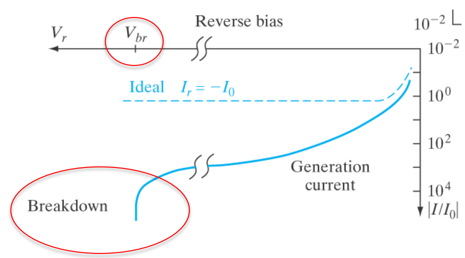
Reverse Bias Non-Idealities

5. Zener Breakdown (quantum mechanical tunneling) Sec. (5.4.1)





- An electron in the valence band of the p-side tunnels through the bandgap (Zener tunneling).
- Need narrow W ➔ heavy doping. High N_A AND N_D.
- V_{br} < 4V often Zener breakdown.

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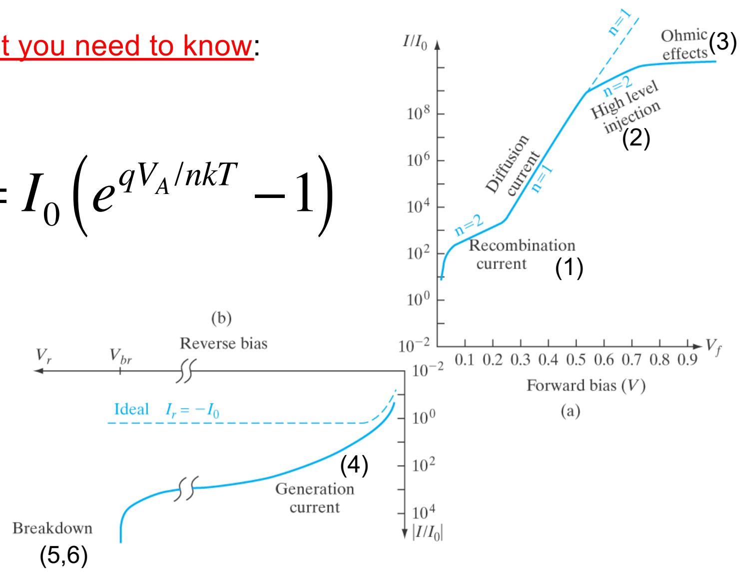
Summary Non-Idealities

What you need to know:

$$I = I_0\left(e^{qV_A/nkT} - 1\right)$$



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Summary Ideal Diode Equation

1. Ideal diode current. Also called the diffusion current. Calculated from the minority carrier diffusion equation at each edge of the depletion region. $I = I_0\left(e^{qV_A/kT} - 1\right)$
2. Law of the junction: Anywhere in the depletion region, $np = n_i^2 e^{qV_A/kT}$.
3. At the p-side edge of the depletion region, $n_p(-x_p) = \frac{n_i^2}{N_A} e^{qV_A/kT} = n_{p0} e^{qV_A/kT}$ giving $\Delta n_p(-x_p) = \frac{n_i^2}{N_A}\left(e^{qV_A/kT} - 1\right)$.
4. At the n-side edge of the depletion region, $p(x_n) = \frac{n_i^2}{N_D} e^{qV_A/kT} = p_{n0} e^{qV_A/kT}$ giving $\Delta p_n(x_n) = \frac{n_i^2}{N_D}\left(e^{qV_A/kT} - 1\right)$

$$I_0 = q\, n_i^2 \left[\frac{D_n}{N_A\, L_n} + \frac{D_p}{N_D\, L_p}\right] = q\, n_i^2 \left[\frac{1}{N_A}\sqrt{\frac{D_n}{\tau_n}} + \frac{1}{N_D}\sqrt{\frac{D_p}{\tau_p}}\right] = q\left[n_{p0}\frac{L_n}{\tau_n} + p_{n0}\frac{L_p}{\tau_p}\right]$$

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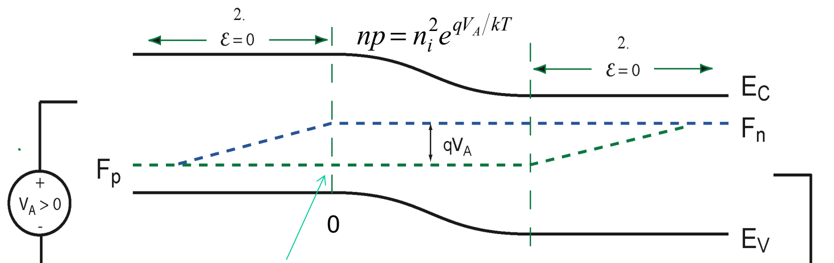
LAW OF THE JUNCTION

General case in non-equilibrium: $np = n_i^2 e^{(F_n - F_p)/kT}$



$p_p(0) = N_A$ Low level injection.

$n_p N_A = n_i^2 e^{qV_A/kT}$ Law of the junction applied at the junction edge.

$$n_p(0) = \frac{n_i^2}{N_A} e^{qV_A/kT} = n_{p0} e^{qV_A/kT}$$

For MCDE, need δn_p:

$$\delta n_p(0) = \frac{n_i^2}{N_A} e^{qV_A/kT} - \frac{n_i^2}{N_A} = \frac{n_i^2}{N_A}\left(e^{qV_A/kT} - 1\right)$$

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Example

- Calculating diode current:
  - On p-side: $N_A = 10^{17}$ cm^{-3}, $\tau_n = 0.1$ μs, $\mu_p = 200$ cm^2/Vs, $\mu_n = 700$ cm^2/Vs.
  - On n-side: $N_D = 10^{15}$ cm^{-3}, $\tau_p = 10$ μs, $\mu_p = 400$ cm^2/Vs, $\mu_n = 1000$ cm^2/Vs.

$$I_0 = q\, n_i^2 \left[\frac{D_n}{N_A L_n} + \frac{D_p}{N_D L_p}\right] = q\, n_i^2 \left[\frac{1}{N_A}\sqrt{\frac{D_n}{\tau_n}} + \frac{1}{N_D}\sqrt{\frac{D_p}{\tau_p}}\right] = q\left[n_{p0}\frac{L_n}{\tau_n} + p_{n0}\frac{L_p}{\tau_p}\right]$$

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$$I_0 = q\, n_i^2 \left[\frac{1}{N_A}\sqrt{\frac{D_n}{\tau_n}} + \frac{1}{N_D}\sqrt{\frac{D_p}{\tau_p}} \right]$$

How do we get D_p and D_n?

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Example

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  - On p-side: $N_A = 10^{17}$ cm^{-3}, $\tau_n = 0.1$ μs, $\mu_p = 200$ cm^2/Vs, $\mu_n = 700$ cm^2/Vs.
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$$I_0 = q\, n_i^2 \left[\frac{1}{N_A}\sqrt{\frac{D_n}{\tau_n}} + \frac{1}{N_D}\sqrt{\frac{D_p}{\tau_p}} \right]$$

Use Einstein relation: $D/\mu = kT/q$.

$$I_0 = q\, n_i^2 \left[\frac{1}{N_A}\sqrt{\frac{\mu_n kT}{\tau_n q}} + \frac{1}{N_D}\sqrt{\frac{\mu_p kT}{\tau_p q}} \right] = \sqrt{qkT}\, n_i^2 \left[\frac{1}{N_A}\sqrt{\frac{\mu_n}{\tau_n}} + \frac{1}{N_D}\sqrt{\frac{\mu_p}{\tau_p}} \right]$$

Which μ_n and μ_p?

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Example

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Which μ_n *and* μ_p? D_n and D_p are the diffusion coefficients of <u>minority</u> carriers.
Use μ_n on the p-side and μ_p on the n-side.

37

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Which μ_n *and* μ_p? D_n and D_p are the diffusion coefficients of <u>minority</u> carriers.
Use μ_n on the p-side and μ_p on the n-side.

$$I_0 = q\, n_i^2 \sqrt{\frac{kT}{q}} \left[\frac{1}{N_A}\sqrt{\frac{\mu_n}{\tau_n}} + \frac{1}{N_D}\sqrt{\frac{\mu_p}{\tau_p}} \right] = (1.602e-19)\; 10^{20}\; \sqrt{(0.026)} \left[\frac{1}{10^{17}}\sqrt{\frac{700}{10^{-7}}} + \frac{1}{10^{15}}\sqrt{\frac{400}{10^{-5}}} \right]$$

$$I_0 = 1.85\times 10^{-11}\ \text{A/cm}^2 = 18.5\ \text{pA/cm}^2$$

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Example

- Calculating diode current:
  - On p-side: $N_A = 10^{17}$ cm^{-3}, $\tau_n = 0.1$ μs, $\mu_p = 200$ cm^2/Vs, $\mu_n = 700$ cm^2/Vs.
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$$I_0 = 1.85\times10^{-11}\ \text{A/cm}^2 = 18.5\ \text{pA/cm}^2$$

$$I = 1.85\times10^{-11}\ \left(e^{qV_A/kT} - 1\right)\ (\text{A/cm}^2)$$

$$V_{bi} = \frac{kT}{q}\ln\left(\frac{N_A N_D}{n_i^2}\right) = 0.026\ln\left(\frac{10^{32}}{10^{20}}\right) = 0.718\ \text{V}$$

$$I(V_A = 0.6) = 1.85\times10^{-11}\ \left(e^{0.6/0.026} - 1\right) = \ 0.195\ \ (\text{A/cm}^2)$$

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Example – Unit Check

$$I_0 = q\ n_i^2\ \left[\frac{1}{N_A}\sqrt{\frac{D_n}{\tau_\text{n}}} + \frac{1}{N_D}\sqrt{\frac{D_p}{\tau_\text{p}}}\ \right] \Rightarrow \frac{\text{C}}{\text{cm}^6}\left[\text{cm}^3\sqrt{\frac{\text{cm}^2}{\text{s}\cdot\text{s}}}\right] = \frac{\text{C}}{\text{s}\cdot\text{cm}^2} = \frac{\text{A}}{\text{cm}^2}$$

$$I_0 = q\ n_i^2\ \sqrt{\frac{kT}{q}}\left[\frac{1}{N_A}\sqrt{\frac{\mu_n}{\tau_\text{n}}} + \frac{1}{N_D}\sqrt{\frac{\mu_p}{\tau_\text{p}}}\ \right] \Rightarrow \frac{\text{C}}{\text{cm}^6}\sqrt{\text{V}}\left[\text{cm}^3\sqrt{\frac{\text{cm}^2}{\text{V}\cdot\text{s}^2}}\right] = \frac{\text{C}}{\text{cm}^2\cdot\text{s}} = \frac{\text{A}}{\text{cm}^2}$$

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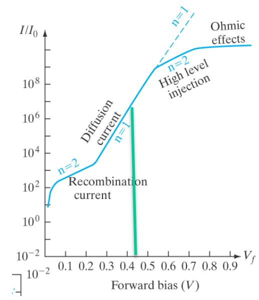
Example2 – Temperature Dependence

- Given V_A lies in the ideal n=1 region, what happens to the current as the temperature is reduced?

$$I = I_0\left(e^{qV_A/kT} - 1\right)$$

$$J = q\, n_i^2 \left[\frac{D_n}{N_A\, L_n} + \frac{D_p}{N_D\, L_p}\right]\left(e^{qV_A/kT} - 1\right)$$



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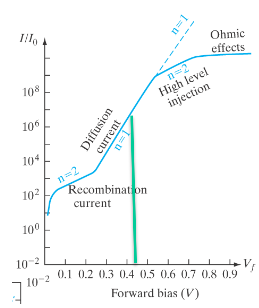

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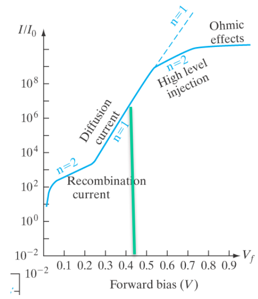
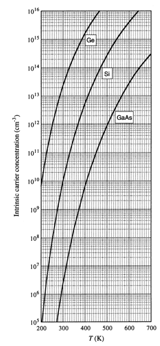
$$n_i = \sqrt{N_C N_V}\;\; e^{-E_G/(2k_BT)}$$





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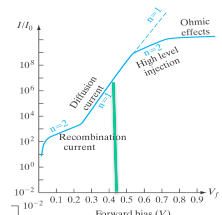
$$J = q\, n_i^2 \left[\frac{D_n}{N_A\, L_n} + \frac{D_p}{N_D\, L_p}\right]\left(e^{qV_A/kT} - \cancel{1}\right) \quad \text{(Forward bias)}$$

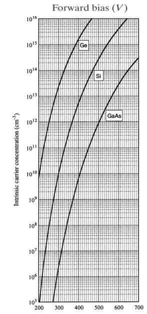
$$n_i^2 = N_C N_V e^{-E_G/(k_BT)} = 4\left(m_n^* m_p^*\right)^{3/2}\left(\frac{k_B}{2\pi\hbar^2}\right)^3 T^3 e^{-E_G/(k_BT)}$$

$$J = q\, N_C N_V \left[\frac{D_n}{N_A\, L_n} + \frac{D_p}{N_D\, L_p}\right] e^{-(E_G - qV_A)/kT}$$

$J \propto T^3 e^{-(E_G - qV_A)/(k_BT)}$ $\quad (3k_BT < V_A < E_G)$ $\quad$ Forward bias

$J \propto T^3 e^{-E_G/(k_BT)}$ $\quad (V_A < -3k_BT)$ $\quad$ Reverse bias





1. Intrinsic carrier concentration n_i decreases exponentially as T decreases.
2. Current **decreases** as T decreases.

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Summary Ideal Diode Equation

1. Ideal diode current. Also called the diffusion current. Calculated from the minority carrier diffusion equation at each edge of the depletion region. $I = I_0\left(e^{qV_A/kT} - 1\right)$
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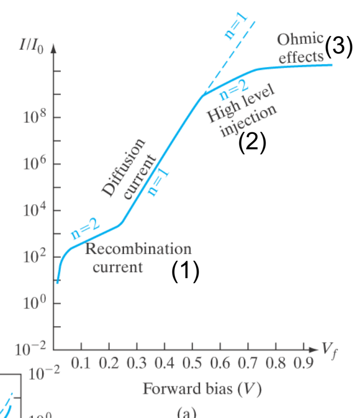
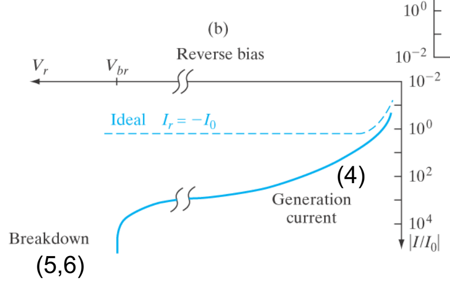
Summary Non-Idealities

What you need to know:

$$I = I_0\left(e^{qV_A/nkT} - 1\right)$$





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